

Customer Lifetime Value Segmentation Report

An end-to-end machine learning analysis combining BG/NBD probabilistic purchase modelling, Gamma-Gamma monetary value estimation, and K-Means customer clustering to surface actionable segments and revenue forecasts.

Report Date: May 23, 2026

Executive Overview

This report summarises the outputs of the CLV segmentation pipeline. Customers were grouped into behavioural segments using probabilistic purchase and monetary models. The key metrics below provide a high-level view of cohort health, predicted value, and model accuracy.

TOTAL CUSTOMERS	AVG PREDICTED CLV	NO. OF SEGMENTS	SILHOUETTE SCORE
4,338	\$89.04	2	0.973

CLV Model Evaluation

The BG/NBD model was trained on an 80% temporal training window and evaluated on the remaining 20% holdout period. Predictions for expected purchases and revenue were compared against actual observed values.

Metric	Value
BG/NBD Log-Likelihood	1.4647
Gamma-Gamma Log-Likelihood	-8.6045
Holdout Period (days)	74
Purchases MAE	0.8349
Purchases RMSE	1.3541
Revenue MAE	\$606.36
Revenue RMSE	\$3,440.18

Clustering Evaluation

Internal clustering validity indices assess the quality of customer segments. A higher Silhouette Score and Calinski-Harabasz Index, along with a lower Davies-Bouldin Index, indicate well-separated, compact clusters.

Index	Score
Silhouette Score	0.9728
Davies-Bouldin Index	0.0193
Calinski-Harabasz Index	1691.44

Segment Profiles

Each customer segment is profiled by its average behavioural and financial characteristics. Recency = days since last purchase; Frequency = unique purchase dates; Monetary = average transaction value.

Segment	Customers	Avg Recency	Avg Freq	Avg Monetary	Avg CLV
High Value Customers	1	1.0d	2.0	\$84,236.25	\$45,663.74
Low Value / Lost	4337	92.6d	3.9	\$399.84	\$78.53

Strategic Recommendations

The following recommendations are derived from the actual cluster statistics of this dataset. Each segment is assigned a strategic tier based on its average CLV, recency, and purchase frequency relative to the other segments.

High Value Customers | 1 customers | Avg CLV: \$45,663.74

Protect revenue concentration and deepen wholesale relationships

High-value customers -- often wholesale accounts or bulk buyers -- represent a disproportionate share of total revenue. A single account in this tier may generate more revenue than hundreds of retail customers combined. Losing even one is a material business risk.

Strategic focus areas:

1. Dedicated account management: Assign a named account manager to each wholesale relationship. Quarterly business reviews should cover volume forecasts, pricing structures, and upcoming product roadmaps.
2. Tiered procurement pricing: Implement volume-based discount ladders (e.g. 5% at 100 units, 12% at 500 units, 20% at 1,000+ units) to incentivise larger order sizes and lock in forward commitments.
3. Exclusive early access: Offer pre-launch product access, private inventory reservations, and priority fulfilment SLAs to reinforce the premium nature of the wholesale relationship.
4. Retention monitoring: Flag any account whose order frequency drops below its 90-day baseline. Trigger an outreach call within 48 hours before the lapse becomes a churn event.

Low Value / Lost | 4,337 customers | Avg CLV: \$78.53

Programmatic re-engagement and win-back pipeline

Lost and low-value retail customers have either lapsed entirely or never developed a repeat-purchase habit. The cost of re-engagement must be weighed against the expected lifetime value of a recovered customer.

Strategic focus areas:

1. Automated re-engagement pipeline: Configure a 4-step email automation triggered at 60, 90, 120, and 180 days of inactivity. Each step escalates the offer: content email, 10% discount, 20% discount, final win-back with free shipping.
2. Cart and browse abandonment triggers: For customers who return to the site but do not convert, deploy real-time abandonment emails within 1 hour. These recover 5-15% of abandoned sessions.
3. Win-back promotional triggers: Seasonal moments (Black Friday, end-of-year clearance) provide a natural re-entry point. Segment this cohort into a dedicated promotional list for these events.
4. Budget reallocation: Customers who do not respond after 180 days should be suppressed from standard campaigns. Reallocate that marketing spend toward acquisition of new customers.